

Kaden Trigueiro

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EDUCATION

University of Texas at Tyler

Tyler, TX

Bachelor of Science in Computer Science, Minor in Math (GPA 3.79)

Jan 2025 – Expected December 2027

- UT Tyler Dean's List of Academic Achievement (COB) (2025-2026); National Society of Leadership and Success (NSLS)

Tyler Junior College

Tyler, TX

Associate of Science in Computer Science (GPA 4.0)

Jan 2022 – Dec 2024

- Graduated *Summa Cum Laude* (2024); President's List of Academic Achievement (2023–2024); Phi Theta Kappa Honor Society

EXPERIENCE

Undergraduate Research Fellow

June 2026 – Present

HAMMER Engineering Research Center @ Northwestern University

Evanston, IL

- Designed a digital shadow framework of a 6-DOF UR5e robot arm for operating an English Wheel to form sheet metal, enabling tool-path validation in Genesis-Embodied-AI simulation before execution on physical hardware
- Developed a multi-stage tool-path pipeline converting ZYX Euler-angle coordinates to quaternions and transforming them into the robot's planning frame; executed three 300+ waypoint paths via inverse kinematics with per-joint position/force telemetry logging
- Modeled 9 sheet-metal workpieces under three material regimes: rigid-body, FEM, and MPM elasto-plastic to compare deformation behaviors
- Working to translate current environmental setup to OpenUSD for cross-compatibility with Issac-Sim

AI Solutions Apprentice

Sep 2025 – May 2026

University of Texas at Tyler

Tyler, TX

- Contributed to an open-source desktop stop-motion animation application (200+ users) using React, Electron, and JavaScript, leveraging generative AI tools to accelerate feature delivery
- Maintained application stability and code quality by fixing community-reported bugs and implementing user-requested feature enhancements
- Completed certifications focused on the ethical deployment and best practices of generative AI technologies under faculty mentorship

Undergraduate Research Fellow

June 2025 – Aug 2025

National Center for Supercomputing Applications @ UIUC

Urbana-Champaign, IL

- Engineered and trained machine learning models using Keras and TensorFlow to accelerate TD-DFT atomic simulations
- Extended simulation optimization beyond traditional baselines by implementing time-series Deep Neural Networks, including LSTMs and Transformers
- Deployed optimized models into a free, public-facing research web application to broaden accessibility for the scientific community
- Awarded formal recognition by NCSA Director William Gropp for outstanding technical contributions to the summer research program

Certified BOH Trainer (Crew)

Sep 2024 – May 2026

Chipotle Mexican Grill

Tyler, TX

- Ensured proper food and environment safety and tested flavor for consistent customer experience
- Aided in teaching procedure, store standards, and food requirements training for 15+ grill, line, and prep staff
- Contributed to store culture that surpassed all quarterly store metric goals 5x

PROJECTS

Pickleball Shot Classifier

COSC4351

Class Project

UT Tyler

- Led a team in engineering a hybrid Dense and Recurrent Neural Network (RNN) to classify pickleball strokes (forehand, backhand, serve, neutral)
- Utilized TensorFlow Lite MoveNet to extract 13 key skeletal joint positions from video data for feature engineering
- Trained model using a Kaggle P100 GPU, achieving a 0.831 classification accuracy over 150 epochs with 5-fold validation

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, HTML/CSS, MySQL, Bash

Frameworks & Libraries: React, Electron, TensorFlow, PyTorch, Matplotlib, NumPy

Tools & Systems: Git/GitHub, Linux, MATLAB, Excel, Overleaf, Blender

Core Skills: Deep Learning, Time Series Analysis, 3D Modeling, Data Science, Statistics, Research, Verbal & Written Communication